Taplend Loan Prediction

Project summary and modelling workflow

Objective

The project combines customer application details with historical loan outcomes to estimate whether a loan is likely to become bad. The prediction can support review of future applications, but it should not be treated as the only basis for a lending decision.

1. Load and prepare the data

- Read the **Application Data** and **Loan Performance** sheets from the Excel workbook.
- Extract the customer identifier and loan number from `idLoan`.
- Standardize identifiers and keep the first loan where the notebook's logic requires it.
- Merge the two tables and create the target label: 1 for a bad loan and 0 for a good loan.
- Check missing values, duplicate customers, status values, and the distribution of outcomes.

2. Data quality and exploratory analysis

The notebook compares matched and unmatched loan records and checks whether the two groups have different bad-loan rates. It also examines outcome rates by selected categories, numerical feature distributions, FICO score correlations, and changes in bad-loan rate by application month.

3. Feature preparation

- Calculate age at application and ratios such as rent-to-income and loan-to-income.
- Treat zero income as missing before calculating ratios.
- Treat the FICO value 222 as a missing-score code and calculate an average FICO feature.
- Group some loan-purpose labels and create a feature indicating whether another phone is listed.
- Exclude identifiers and fields that should not be used as predictors.

4. Model training

The notebook uses a stratified 80:20 train-test split. A preprocessing pipeline imputes missing numerical values with the median, imputes categorical values with the most frequent category, scales numerical features, and one-hot encodes categorical features.

- Logistic Regression is used as a baseline.
- Random Forest and Extra Trees are compared with the baseline.
- Five-fold stratified cross-validation compares ROC-AUC on the training data.

5. Evaluation and approval cutoff

The test set is evaluated with ROC-AUC, average precision, accuracy, precision, recall, F1-score, and the KS statistic. ROC curves and Logistic Regression coefficients are also reviewed.

The notebook then checks several probability cutoffs. Applicants with predicted bad-loan probability below the selected cutoff are treated as approved in the illustrative policy analysis. A lower cutoff is stricter and generally approves fewer applicants.

6. Saving the model

The selected Logistic Regression pipeline is fitted on the available labelled data and saved as `taplend_loan_model.joblib`. The saved pipeline includes preprocessing and the classifier so that the same transformations can be applied to new applications.

Limitations and considerations

Data coverage

- Some loan-performance records do not match an application record. These records cannot be used directly for supervised training after the merge.
- The matched and unmatched groups have different observed bad-loan rates, which suggests the training sample may not represent all loans.
- The dataset contains approved applicants only. It does not show outcomes for people who were declined, so the model cannot directly learn how declined applications would have performed.

Model limitations

- The notebook reports that model AUC is only slightly above random guessing. Predictions should therefore be treated as weak signals, not reliable standalone decisions.
- A random train-test split may not reflect future performance as well as a time-based holdout. Application dates should be considered when selecting a validation strategy.
- The cutoff analysis is illustrative. A real lending policy needs explicit costs for bad loans, missed good loans, affordability, and regulatory requirements.

Additional data that could help

- Application outcomes for declined applicants, where legally and ethically available.
- Verified repayment history, arrears, defaults, and time since delinquency.
- Debt obligations, existing credit utilisation, and a reliable measure of disposable income.
- A clear definition of the loan outcome and a consistent observation period for every loan.

Conclusion

The notebook provides an end-to-end baseline workflow for merging loan data, checking data quality, engineering features, comparing classifiers, and saving a prediction pipeline. The current signal appears limited, and the sample-coverage issue is material. More representative data and validation on a later time period are needed before the model is used for lending decisions.

Files and reproducibility

The notebook expects the Excel workbook named `Assessment_-_Data_Scientist.xlsx` to be available in the working directory. Install the notebook's required Python packages before running all cells from top to bottom.